

Probability, Statistics and Data Analysis

4: Continuous and Joint Random Variables

**Absolute certainty is a privilege of
uneducated minds and fanatics.
It is, for scientific folk, an
unattainable ideal.**

Cassius J. Keyser

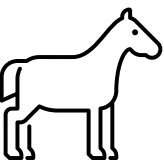
4: Continuous and Joint Random Variables

What we will cover

- Continuous
 - Probability Density Function
 - Cumulative Distribution Function
 - Percentiles
 - Expected Value
 - Variance
- Joint Probability Distributions
 - PMF, PDF and marginals
 - Expected value
 - Independence
- Covariance
- Correlation



Types of Random Variables



Bernoulli: does a soldier survive a kick?

- Outcomes
 - A: Soldier is killed by a horse kick
 - B: Soldier is not killed by a horse kick
- As RV
 - $X(A)=1$
 - $X(B)=0$
 - Bernoulli RV to count number of deaths

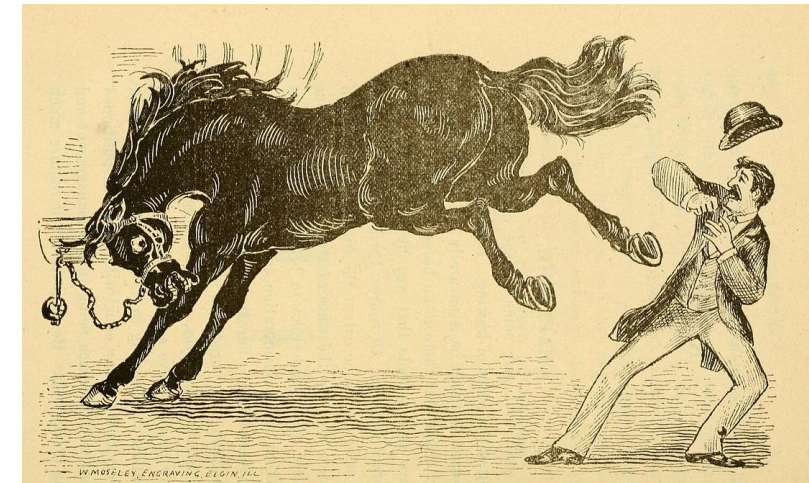
Discrete: number of deaths per year for a given Division

- As RV
 - X = number of deaths in a year for a given division
 - Discrete Random Variable

Continuous: how long does a soldier survive after a kick?

- As RV
 - X = time in hours of survival after being kicked
 - Continuous Random Variable

<https://www.biodiversitylibrary.org/pageimage/20321596>





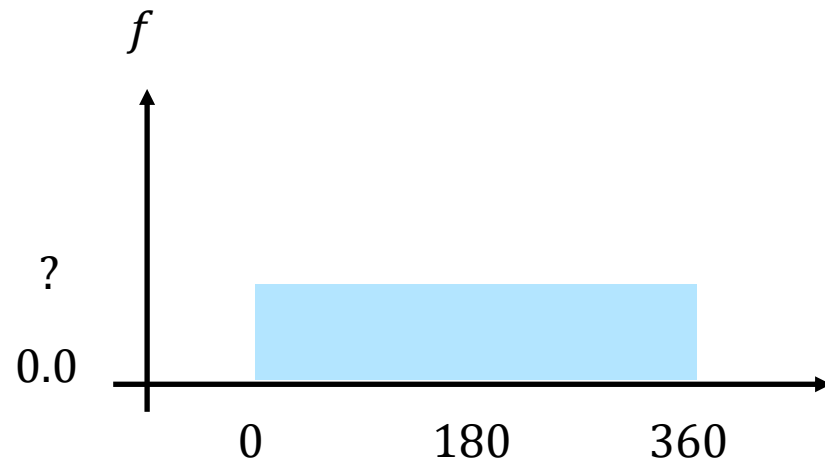
Probability Density Functions

Continuous Probability Distributions

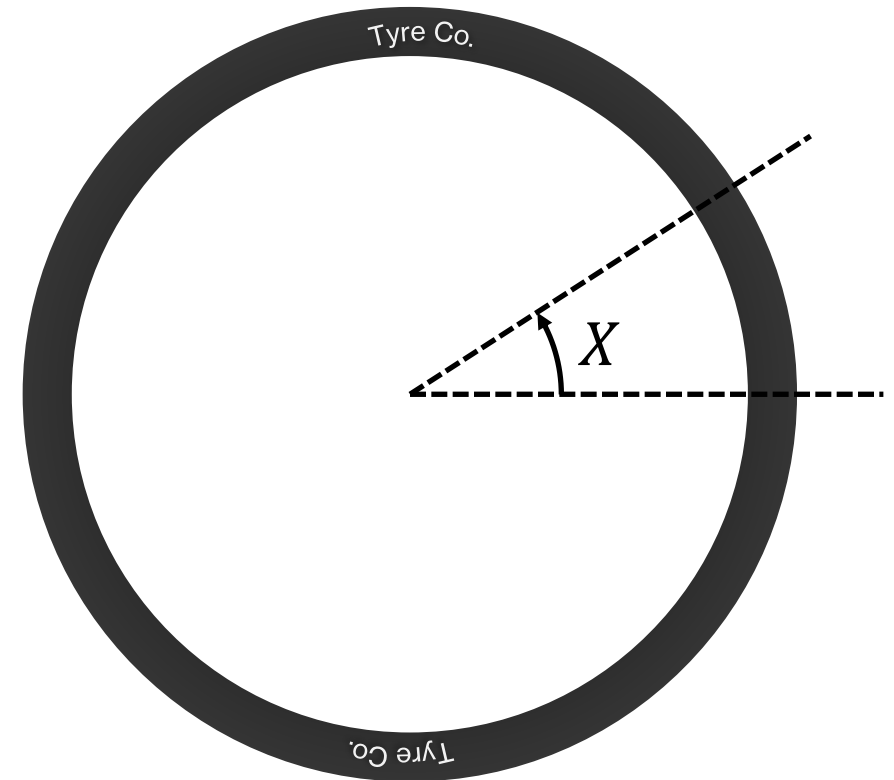
Wheel Damage



- The tire of a bicycle is leaking
- x gives angular location of damage
- Clearly a continuous variable
- With no prior knowledge of location, any value of X is equally likely



$$f(x) = \begin{cases} ? & 0 \leq X < 360 \\ 0 & \text{otherwise} \end{cases}$$



Wheel Damage



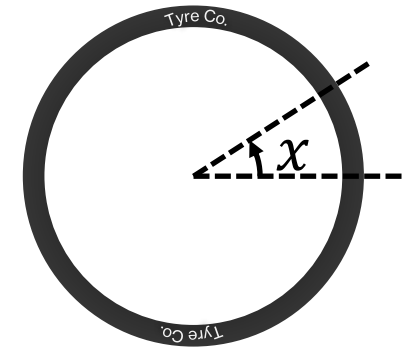
- What is the probability that x is exactly, for example, 10 degrees?

$$P(x = 10^\circ) = 0$$

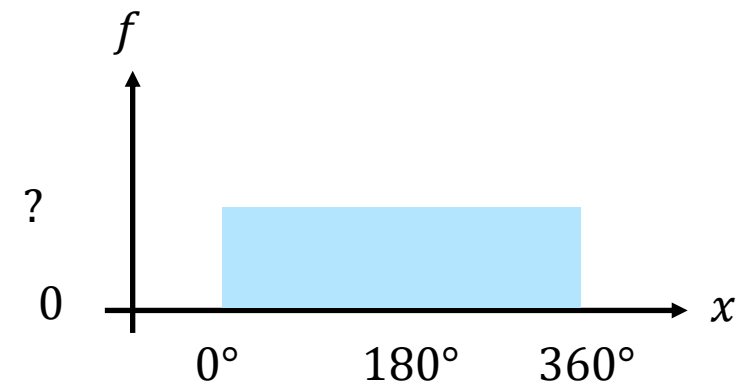
- Probability of exactly 10 is zero, because we could have 10.0000000000000001 which is not 10
- Generally, for continuous RV X

$$P(a \leq X \leq b) = P(a < X \leq b) = \\ P(a \leq X < b) = P(a < X < b)$$

- Real-world relevance may correspond to measurement resolution
 - E.g. if we can measure with resolution of 1 degree then $P(9.5^\circ \leq x \leq 10.5^\circ)$ would be sensible



$$f(x) = \begin{cases} ? & 0^\circ \leq x < 360^\circ \\ 0 & \text{otherwise} \end{cases}$$

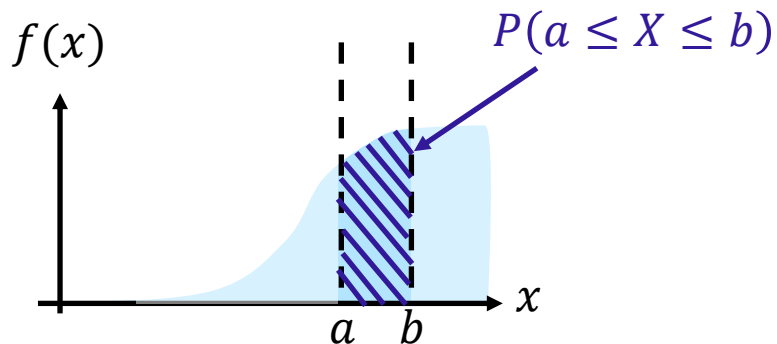


Probability Distributions for Continuous RVs

- For a continuous RV, X , the probability density function $f(x)$ defined according to

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

- The probability that X lies between a and b is the area under $f(x)$ between these two values



Probability Distribution is known as
Discrete RV: Probability Mass Function, PMF
Continuous RV: Probability Density Function, PDF

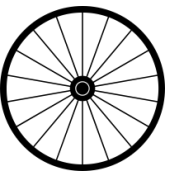
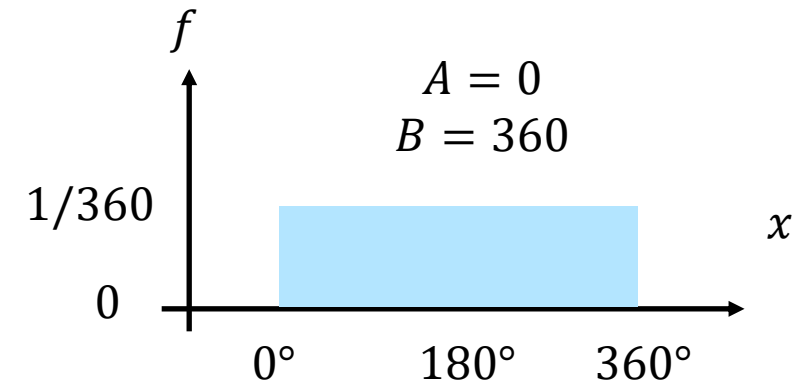
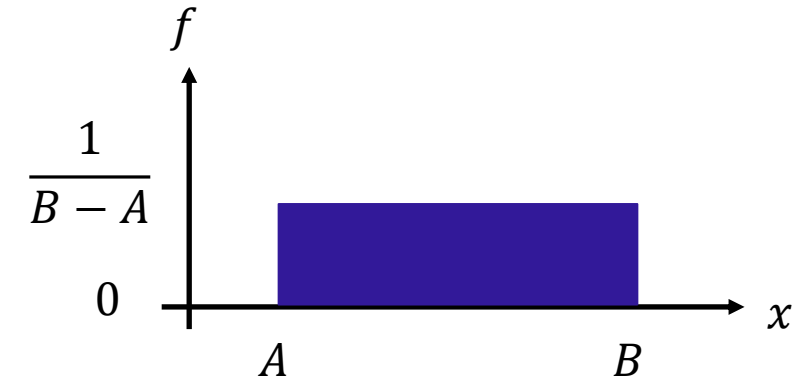
Uniform Distribution

- If a continuous RV is constant in the interval $[A,B]$ it is said to have a Uniform Distribution with PDF

$$f(x; A, B) = \begin{cases} \frac{1}{B - A} & A \leq x < B \\ 0 & \text{otherwise} \end{cases}$$

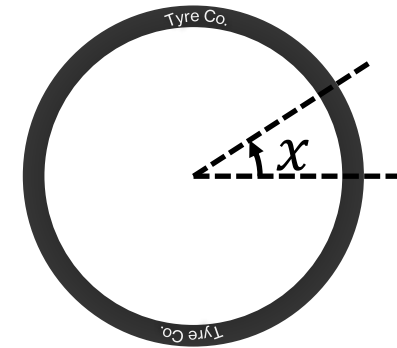
Not on summary sheet, as too simple!

- Effectively, everything is equally likely

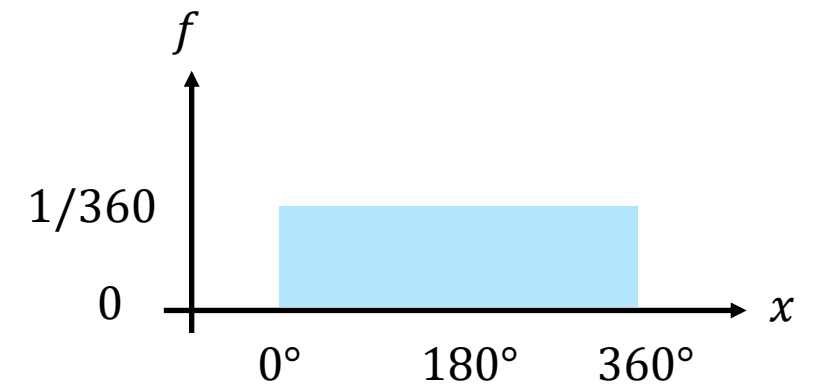


Wheel Damage

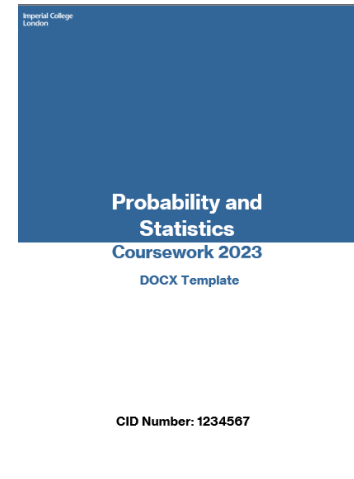
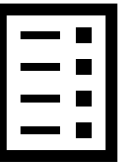
What is the probability that x is between 10 and 20 degrees?



$$f(x) = \begin{cases} 1/360 & 0^\circ \leq x < 360^\circ \\ 0 & \text{otherwise} \end{cases}$$



Relative Work level on Coursework



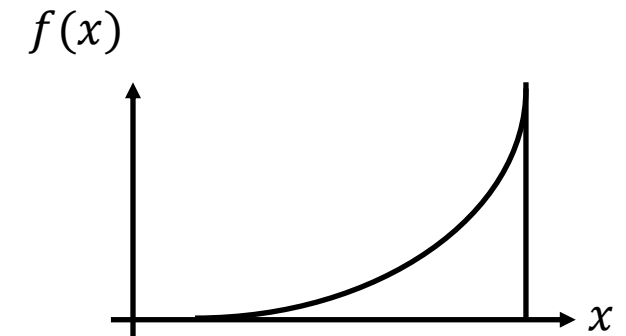
- Let X be the time after the coursework is announced in weeks
- $f(x)$ is the probability density function describing the probability of a student working on the coursework during working hours

- Assume $f(x) = \begin{cases} kx^2 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$

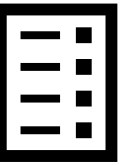
- How do we find k ?
- Use the fact that

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Axiom 2: $P(\mathcal{S}) = 1$



Relative work level as a continuous RV



$$f(x) = \begin{cases} kx^2 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$



Cumulative Distribution Functions

For continuous variables

Cumulative Distribution Functions for Continuous RV

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$$

y is dummy variable. Equivalent notation:

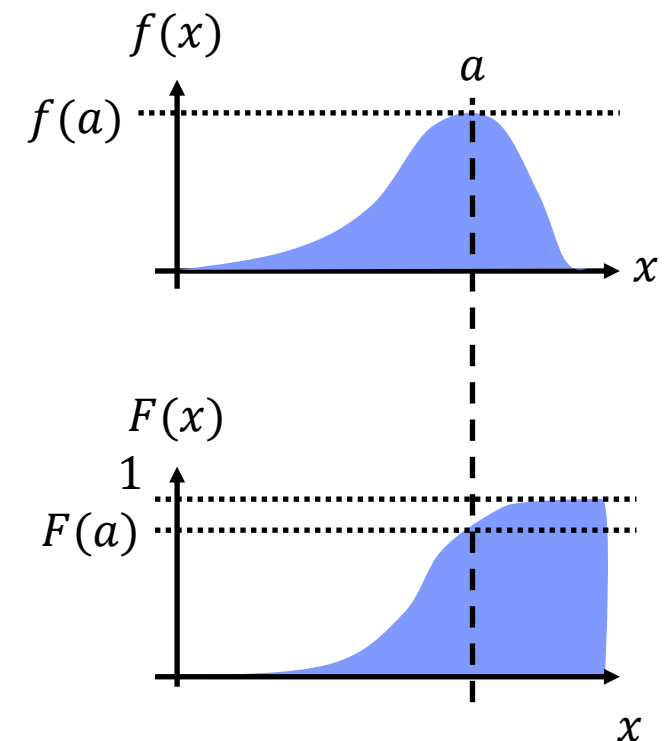
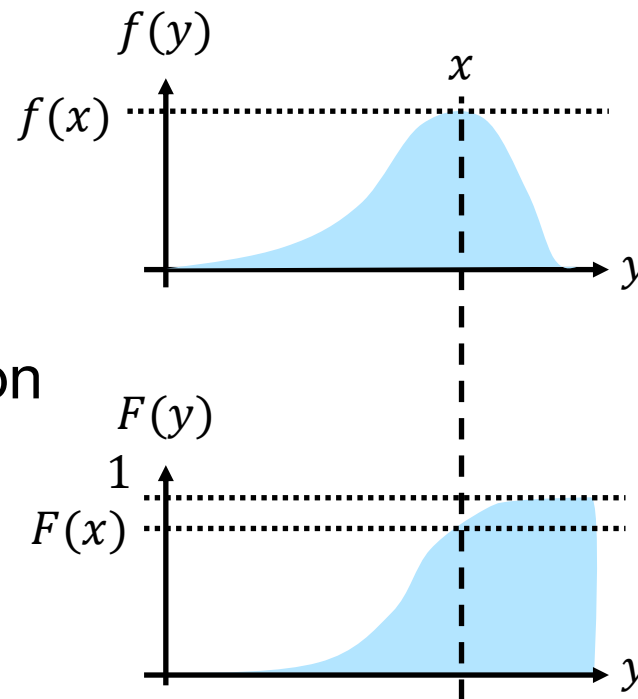
$$F(a) = P(X \leq a) = \int_{-\infty}^a f(x) dx$$

Probability that the RV X is less than or equal to a specific value x

- CDF is integral under the probability distribution function

Discrete, for comparison

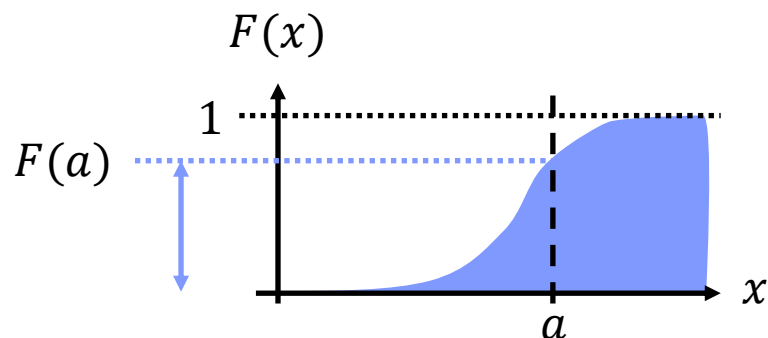
$$F(x) = P(X \leq x) = \sum_{x:y \leq x} p(y)$$



Using CDF for continuous RV

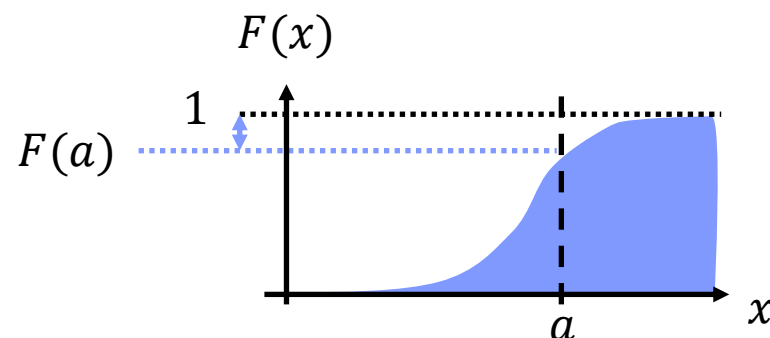
Less than

$$P(X \leq a) = F(a)$$



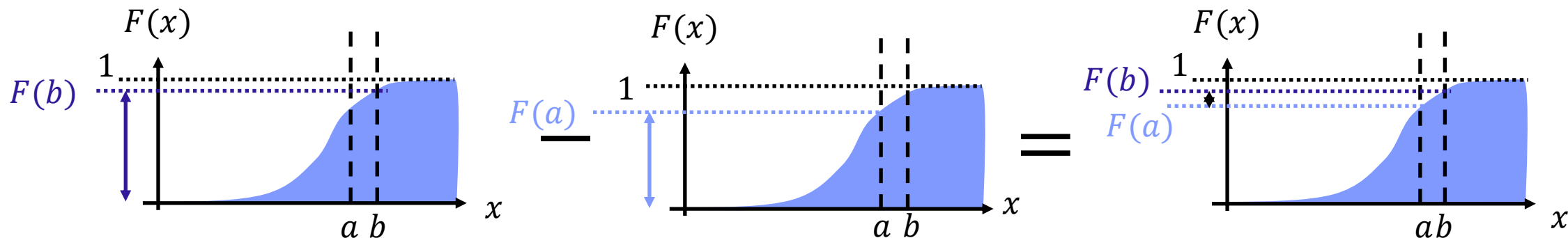
More than

$$P(X \geq a) = 1 - F(a)$$

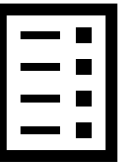


Between

$$P(a \leq X \leq b) = F(b) - F(a)$$



Work level k via CDF



$$f(x) = \begin{cases} kx^2 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$F(a) = \frac{ka^3}{3} \text{ for } 0 \leq x \leq 2$$

First find CDF

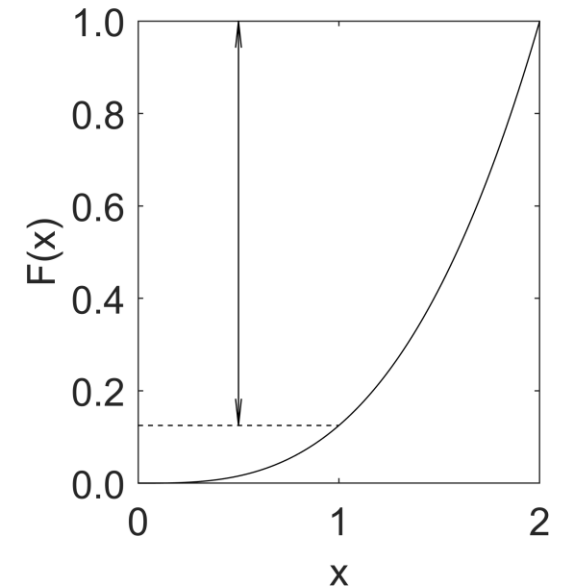
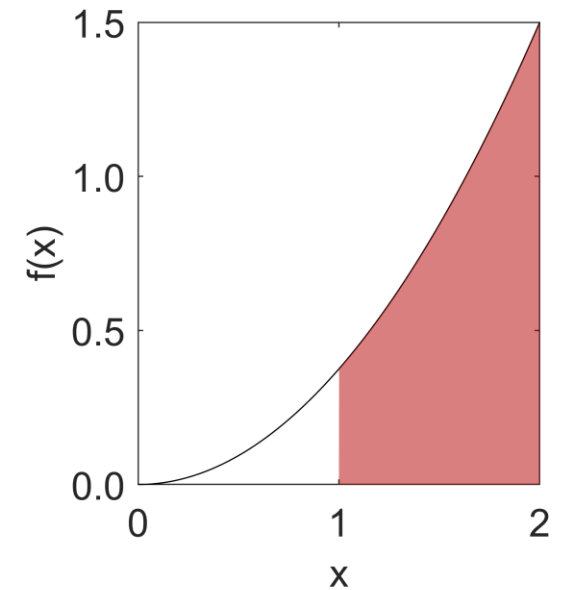
Then find $F(\infty) = 1$

In this problem, $F = 1$ represents the total time spent on the coursework

Work level using CDF

- What proportion of the work is done in the second week?

$$F(a) = \frac{a^3}{8}$$



Hold on!

- $f(x)$ exceeded 1, but probability cannot exceed 1
- Probability density can
- Consider the analogy

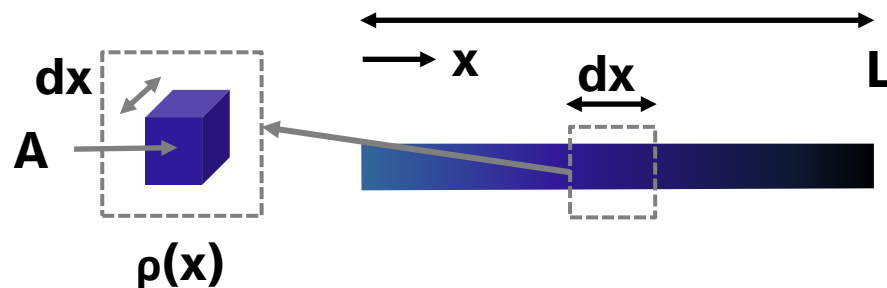
Mass ~ Probability

- For a discrete RV

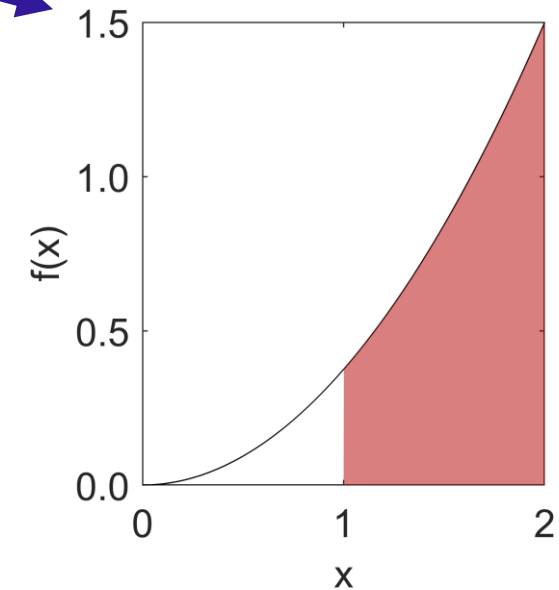


$$m_{total} = \sum_{i=1}^n m_i$$

- For a continuous RV



$$m_{total} = A \int_0^L \rho(x) dx$$



$$F(x) = \sum_{y:y \leq x} p(y)$$

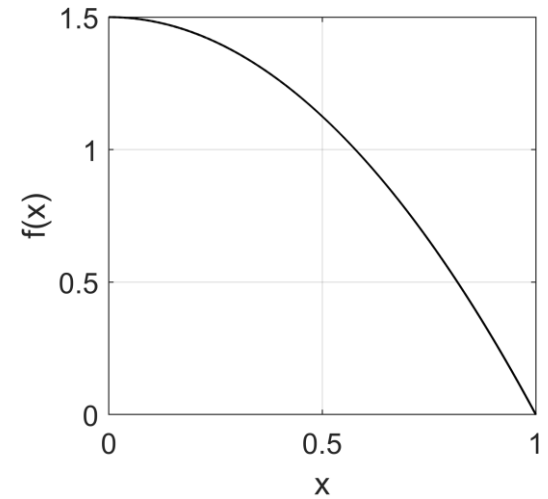
$$F(x) = \int_{-\infty}^x f(y) dy$$

- We would not interpret density as mass!

PDF Validity – C Elegans



- C Elegans are 1mm long self-fertilising hermaphrodites, with life expectancy of approximately 20 days
- They are a common model of rare diseases
- Let x be the relative age of the C Elegans (i.e. age/24 days) at death and $f(x)$ be the PDF describing how likely it is that a given C Elegans will die

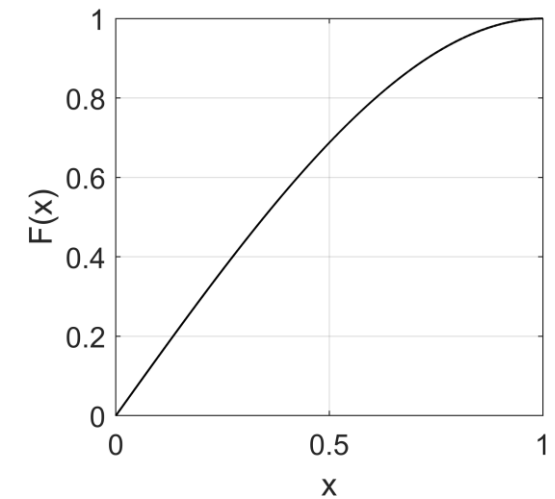
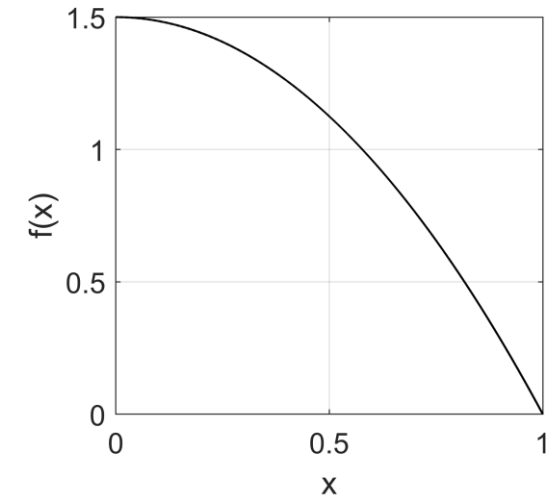


- $$f(x) = \begin{cases} \frac{3}{2}(1 - x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$
- Is this a valid PDF?

PDF validity via CDF



- CDF will represent the probability of being alive
- Find the CDF and hence check the validity of $f(x)$





Percentiles

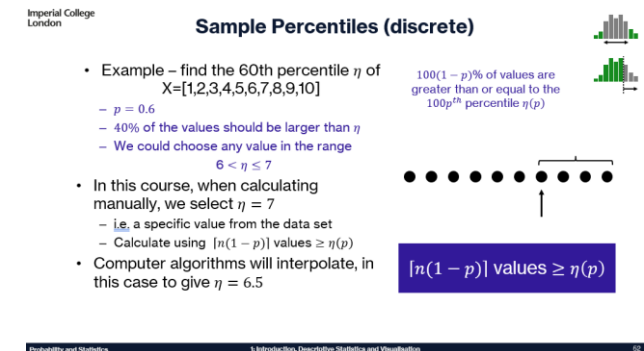
of continuous RV distributions

Continuous Percentiles

- Definition in Lecture 1 given for finite samples
- It was one of up to 9 different options, depending on how we chose to deal with numbers between finite observations
- For continuous RV, we don't have this ambiguity
- For a continuous PDF, the $100p^{th}$ percentile $\eta(p)$ is defined according to

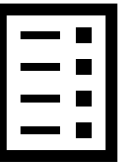
$$p = P(X \leq \eta(p)) = F[\eta(p)] = \int_{-\infty}^{\eta(p)} f(y) dy$$

- It is the CDF with $\eta(p) = a$
- For example, consider the median
 - 50th percentile, $p = 0.5$
 - The median is $\eta(0.5) = \tilde{\mu}$ satisfies $F[\tilde{\mu}] = 0.5$

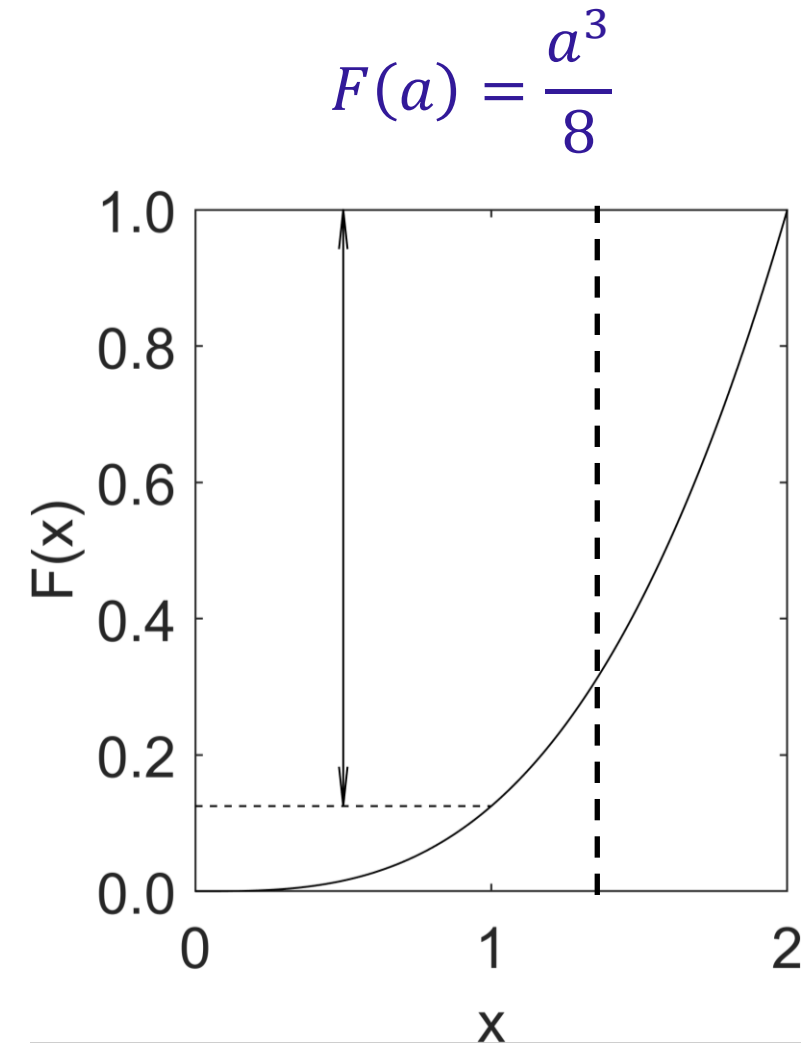


Note that as we move further into statistics we use this a lot, but tend to drop the terminology 'percentiles' in favour of just talking about probabilities

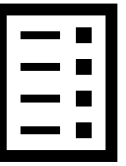
Work level percentiles



- We asked before “What proportion of the work is done in the second week?”
- We chose x , and found $F(x)$ (or with a)
- What if we choose $F(x)$ and want to find x
- Equivalently, choose $F(\eta(p))$ and want to find $\eta(p)$
- E.g. “When is 50% of the work done”



Inverse CDF - work level percentiles



- “When is 50% of the work done”
- What is $\eta(0.5)$?

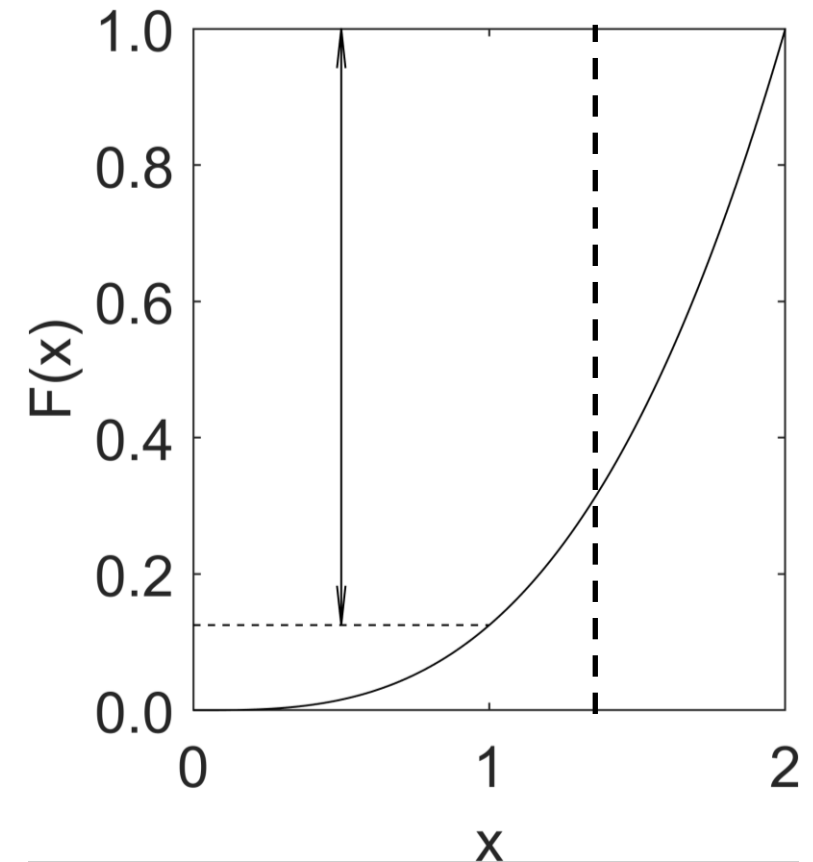
$$F(\eta(p)) = p$$

- Use the inverse

$$\eta(p) = F^{-1}(p)$$

$$F(a) = \frac{a^3}{8} = y$$

$$p = P(X \leq \eta(p)) = F[\eta(p)] = \int_{-\infty}^{\eta(p)} f(y) dy$$





Expected Value and Variance

For Continuous RV

Expected Value and Variance of Continuous RV

Continuous

$$E(X) = \mu_X = \int_{-\infty}^{\infty} x f(x) dx$$

Continuous

$$V(X) = \sigma_X^2 = \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x) dx$$

Discrete

$$E(X) = \mu_X = \sum_{x \in D} xp(x)$$

Discrete

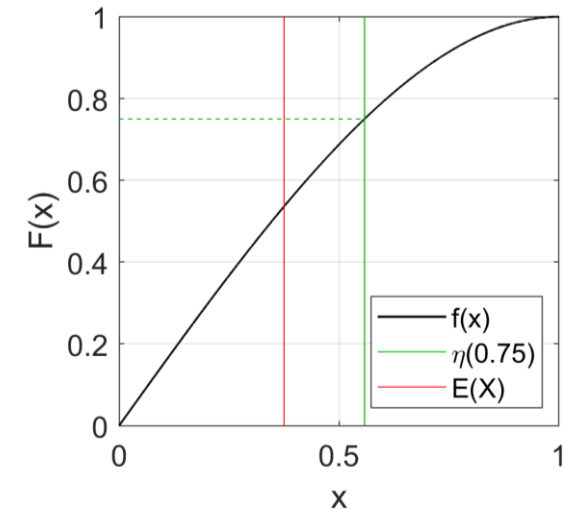
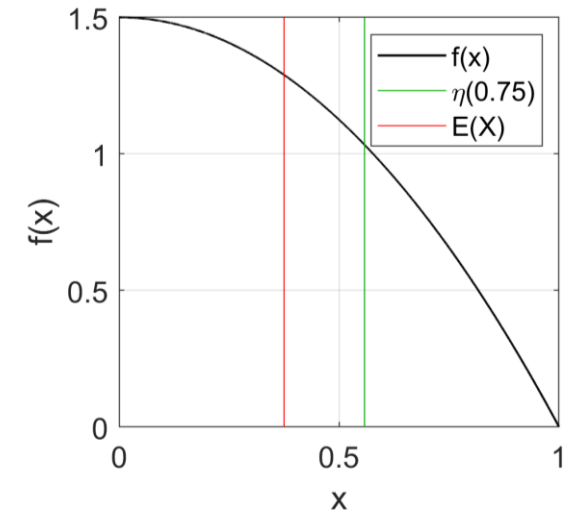
$$V(X) = \sigma_X^2 = \sum_{x \in D} (x - \mu_X)^2 p(x)$$

Expected Value for C Elegans



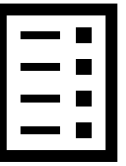
$$f(x) = \begin{cases} \frac{3}{2}(1 - x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$



- We expect the C elegans to die at $3/8 \times 24 = 9$ days ('on average')

Variance for Work level



- Find the variance of the RV X that represents work level on coursework
- First we need E

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

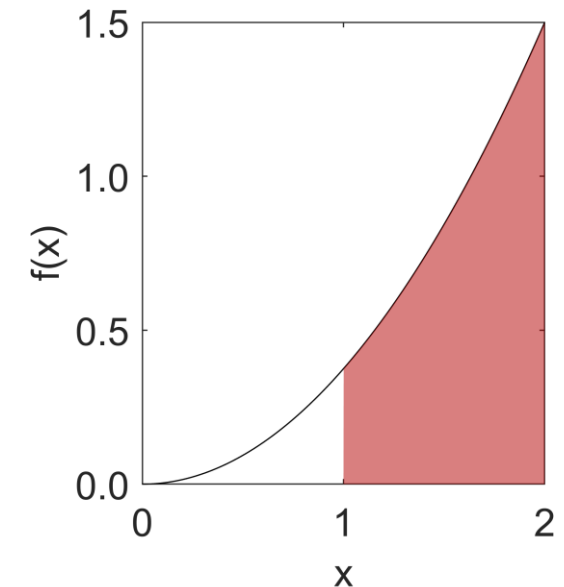
$$f(x) = \begin{cases} \frac{3}{8}x^2 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

- Then

$$V(X) = \int_{-\infty}^{\infty} (x - E(X))^2 f(x) dx$$

- Too much effort for us to solve here. We would use a computer (AI options, Wolfram Alpha, MATLAB symbolic)

$$V(X) = 0.15$$



Differences between CDF and Expected Value



- A quick reminder
- Expected value (and Variance)
 - of an RV X (not a realisation x)

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

- Limits of integration are the whole range of X

- CDF (and percentiles)

- of a specific range (hence lower case x)

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$$

- Limits of integration are lower limit to our specific value

$$f(x) = \begin{cases} \frac{3}{2}(1 - x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \frac{3}{2} \int_0^1 x (1 - x^2) dx$$

$$F(a) = \int_0^a \frac{3}{2} (1 - x^2) dx$$

$$F(x) = \int_0^x \frac{3}{2} (1 - y^2) dy$$



Joint Probability Distributions and RVs

Discrete

PMF for Discrete Joint Random Variables

- So far we have considered only single random variables
 - E.g. the effect of a treatment for diabetes being tested in mice
- What if we are interested in another factor
 - E.g. whether the effect is dependent on the sex of the mice

Let X and Y be two discrete random variables defined on the same sample space $\mathcal{S}$ of an experiment.

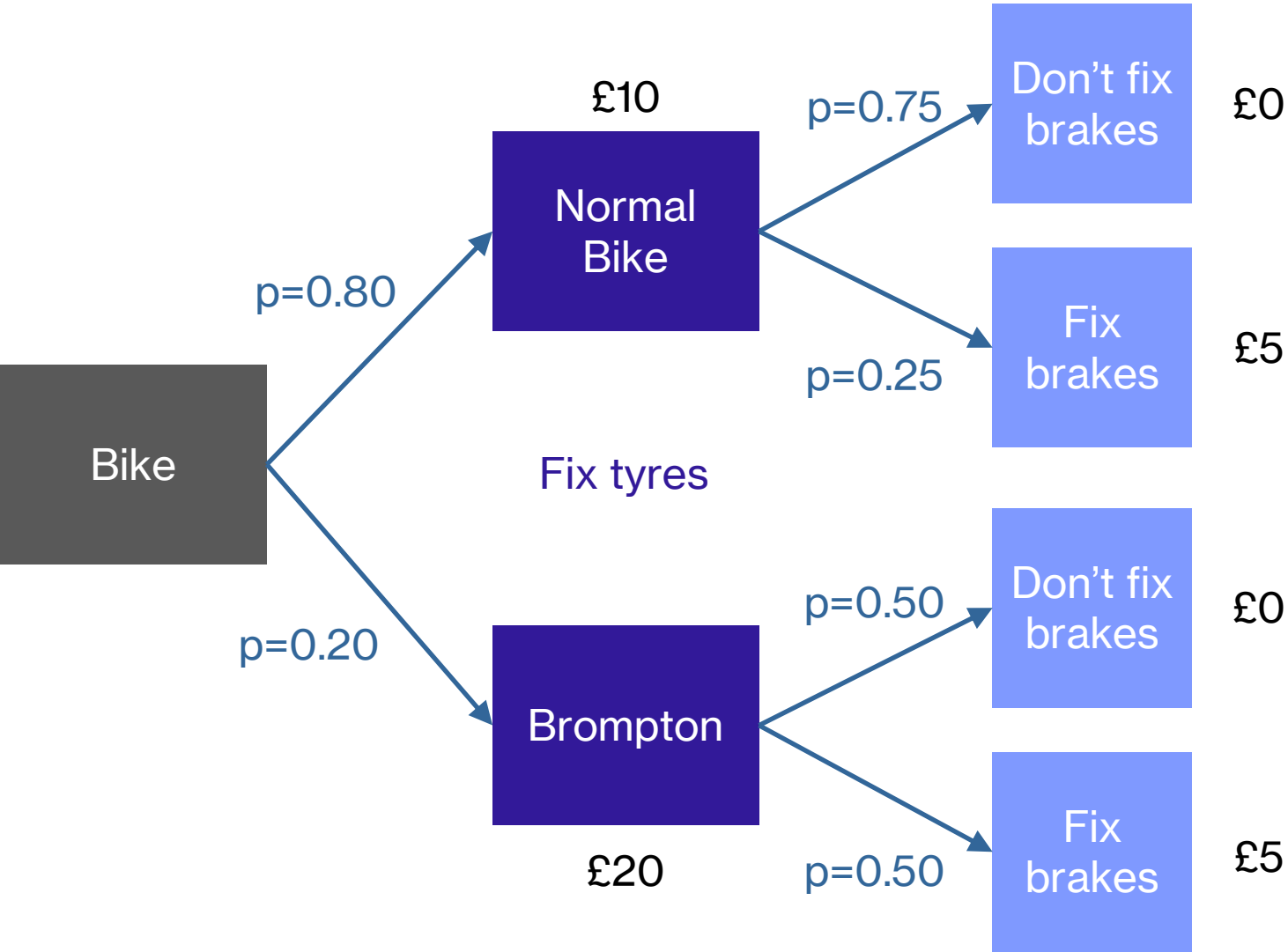
The **joint probability mass function** $p(x, y)$ is defined for each pair of numbers $p(x, y)$ by

$$p(x, y) = P(X = x \text{ and } Y = y)$$

where

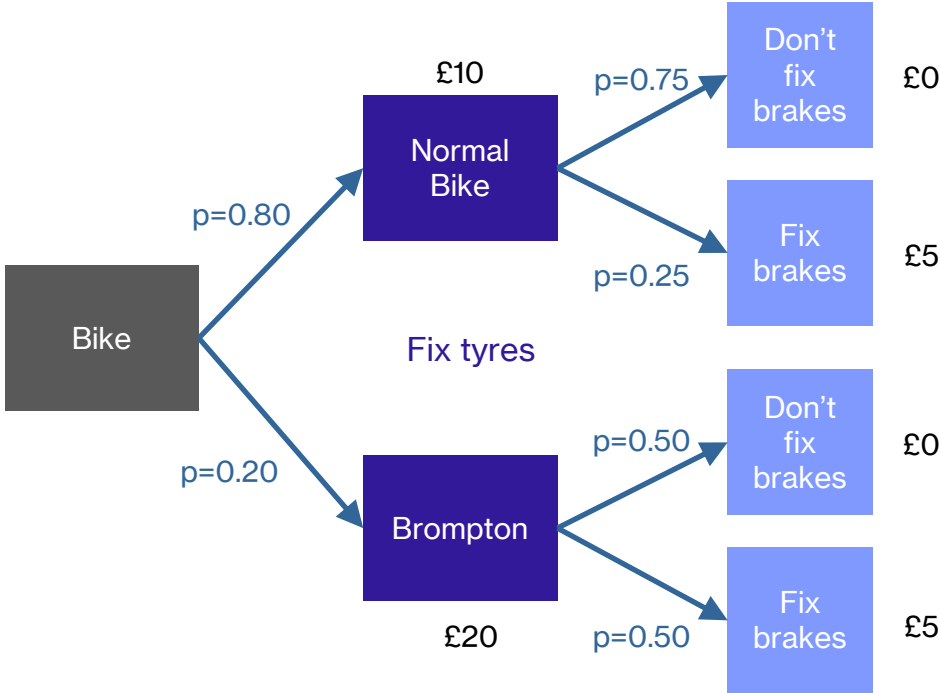
$$\sum_x \sum_y p(x, y) = 1$$

Fixing Tyres and Brakes



<https://vivasport.hr/proizvod/bicikl-basso-monza-tiagra-10-speed-34-50/>
<https://www.condorcycles.com/products/brompton-s2l-2021-folding-bike>

Joint Probability - Fixing Tyres and Brakes



- Let X be the earning from fixing a flat tyre
- Let Y be the earning from fixing the brakes

$$p(x,y) = P(X = x \text{ and } Y = y)$$

X (£)	Y (£)	X+Y (£)	P(X,Y)
10	0		
10	5		
20	0		
20	5		



Joint Probability Table

$p(x,y)$		x	
		10	20
y	5	0.20	0.10
	0	0.60	0.10

For a subset $A \subset \mathcal{S}$ consisting of pairs of x, y the probability $P[(X, Y) \in A]$ is given by

$\in$ is an element of

$$P[(X, Y) \in A] = \sum_{(x,y)} \sum_{\in A} p(x, y)$$



P probability of A
 p probability of a specific realisation of x and y , that belongs in A

Defining Joint RVs – Fixing Bikes

$p(x,y)$		x	
		10	20
y	5	0.20	0.10
	0	0.60	0.10

$$P[(X, Y) \in A] = \sum_{(x,y)} \sum_{\in A} p(x, y)$$

1. Let A be the subset of values for which $X \geq 20$ (with no specification for Y).

2. Let A be the subset of values for which $X + Y$ is a multiple of 10

Expected Value for Joint RVs



- What is the expected fee for a given bike?
- For a single RV, X

$$E(X) = \sum_x xp(x)$$

- This can also be applied to a function of RV, $h(X)$

$$E(h[X]) = \sum_x h(x)p(x)$$

- $X + Y$ is itself an RV, $h(X, Y)$

$$E(X + Y) = \sum_x \sum_y (x + y)p(x, y)$$

$p(x,y)$		x	
		10	20
y	5	0.20	0.10
	0	0.60	0.10

$$E[h(X, Y)] = \sum_x \sum_y h(x, y)p(x, y)$$

Joint Discrete Variables

Marginal PMFs for Discrete RVs



$p(x,y)$		x		
		10	20	
y	5	0.20	0.10	0.3
	0	0.60	0.10	0.7
		p_X	0.8	0.2

$$\sum_y p_Y(y) = 1$$

$$\sum_x p_X(x) = 1$$

Marginal PMFs
must sum to 1

- Marginal probability distributions are the distributions of one variable alone
- For discrete RVs they are calculated according to

$$p_X(x) = \sum_y p(x, y)$$

$$p_Y(y) = \sum_x p(x, y)$$



Independence for Discrete RVs

- The variables in joint probability distributions can be dependent or independent
- If they are independent, then neither variable depends on the other

$p(x,y)$		x		p_Y
		10	20	
y	5	0.20	0.10	0.3
	0	0.60	0.10	0.7
p_X		0.8	0.2	

Generally, X and Y are independent if for every pair x, y

$$p(x, y) = p_X(x)p_Y(y)$$

E.g. $X = 10, Y = 5$

Imperial College
London

Two-Way Independence

Two Events
Independent – A is unaffected by B

or equivalently

$P(A|B) = P(A)$
 $P(A \cap B) = P(A)P(B)$

• Dependent – A is affected by B
 $P(A|B) \neq P(A)$

Examples –

- Select an individual from the class
 - Height > 180 cm and weight < 70 kg (dependent)
- Select an individual from this class and one from another
 - First student is European and second student is European (independent)

Conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



Joint Probability Distributions and RVs

Continuous

PDF for Continuous Joint RVs



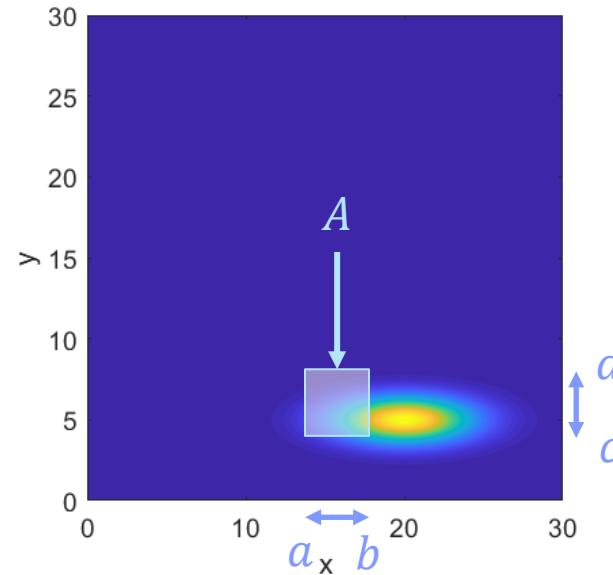
- X : earning from fixing a flat tyre
- Y : earning from fixing the brakes
- Now consider costs are not fixed, but charged at an hourly rate of £20 per hour
- Tyre repair (X) and brakes repair (Y) are now continuous variables

$$P[(X, Y) \in A] = \iint_A f(x, y) dx dy$$

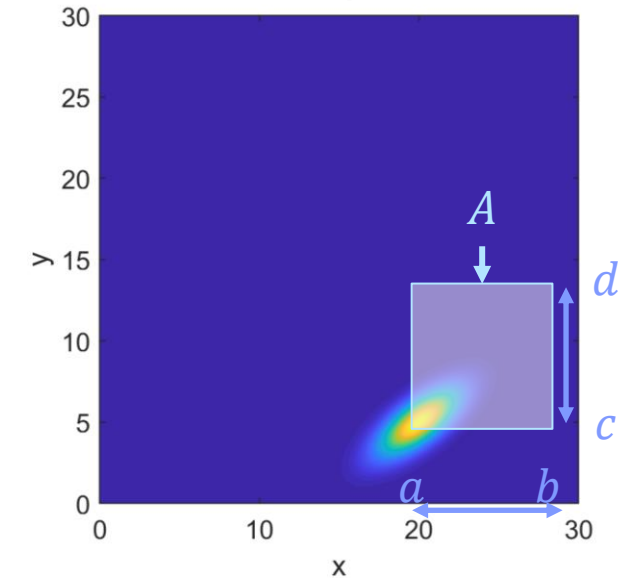
Probability from Probability Density

$$P[a \leq X \leq b, c \leq Y \leq d] = \int_c^d \int_a^b f(x, y) dx dy$$

Example 1



Example 2



Note that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

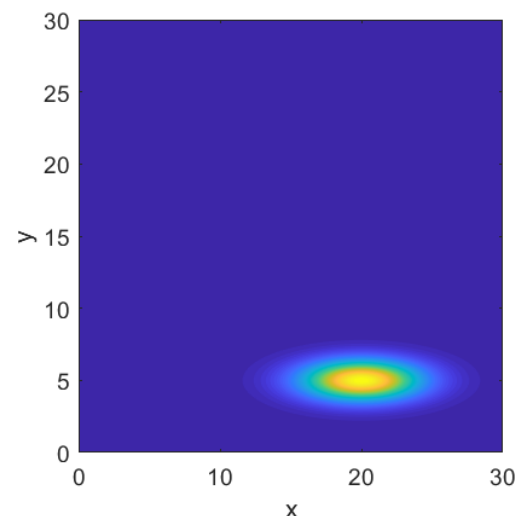
See MATLAB code for how I generated these graphs

Marginal PDFs for Continuous RVs

- As with discrete RVs, we can compute marginal PDFs

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$



Where

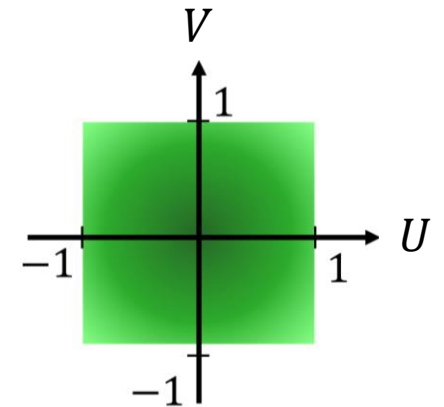
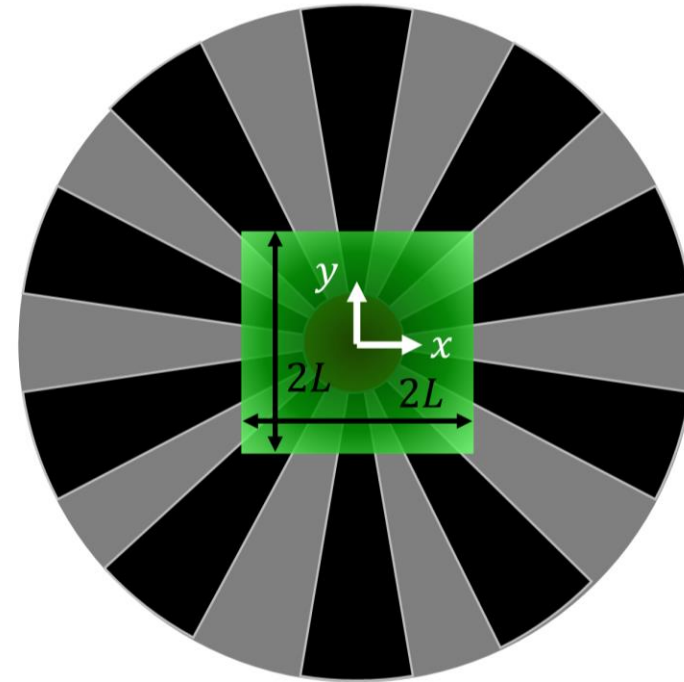
$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$
$$\int_{-\infty}^{\infty} f_Y(y) dy = 1$$

See MATLAB code for
how I generated
these graphs

Dartboard



- A dart player is equally precise with the vertical (y) and horizontal (x) position of their dart placement and they always manage to get the dart in the green square
- Let the RV U be the x co-ordinate of the dart relative to the length L ($u = X/L$). Similarly, let V be the y co-ordinate of the dart relative to the length ($V = Y/L$).
- Considering the centre of the board to be $(0,0)$, the joint probability distribution of U and V is



$$f(u, v) = \begin{cases} \alpha(2 - u^2 - v^2) & \text{if } -1 \leq u \leq 1, -1 \leq v \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Independence for Continuous RVs

- The variables in continuous joint probability distributions can be dependent or independent
- Similarly to discrete variables, this is determined using the marginal PDFs

Generally, X and Y are independent if for all x, y

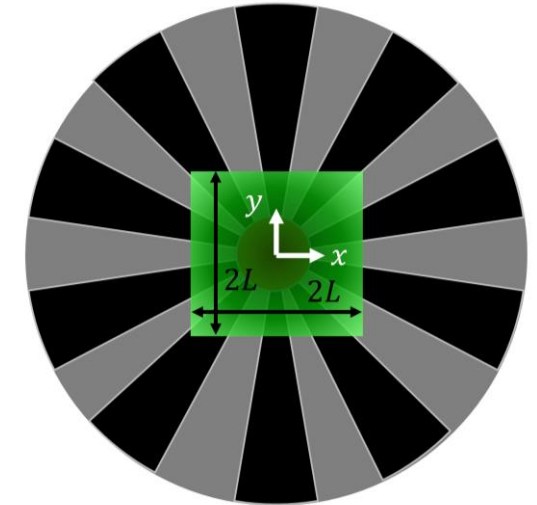
$$f(x, y) = f_X(x)f_Y(y)$$

Dartboard - Marginal PDFs



- Are U and V independent?

Marginal PDF for U



Marginal PDF for V

$$f(u, v) = \alpha(2 - u^2 - v^2)$$

Dartboard – Independence

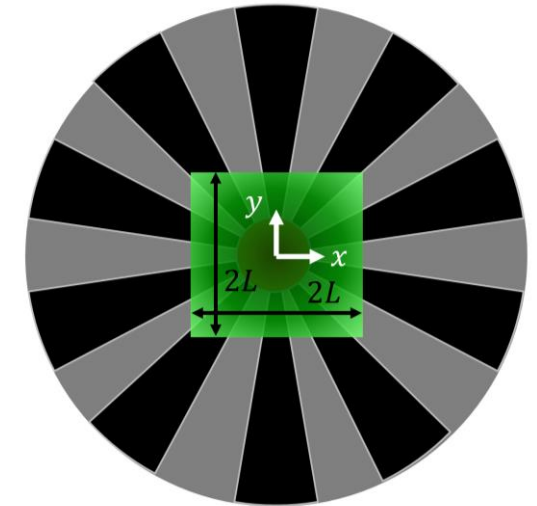


- For independence, $f_U(u)f_V(v) = f(u, v)$

$$f_U(u) = 2\alpha \left(\frac{5}{3} - u^2 \right)$$

$$f_V(v) = 2\alpha \left(\frac{5}{3} - v^2 \right)$$

$$f(u, v) = \alpha(2 - u^2 - v^2)$$



$$f(u, v) = \alpha(2 - u^2 - v^2)$$

Note that $f_U(u)$ and $f_V(v)$ are each only functions of their own variable by definition

This does not mean that u and v are independent

Dartboard – Find α



- Marginal PDFs are just PDFs
- Hence they follow the rule

$$F(\infty) = 1$$

or equivalently

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

- Use this to find α

$$f_U(u) = 2\alpha \left(\frac{5}{3} - u^2 \right)$$

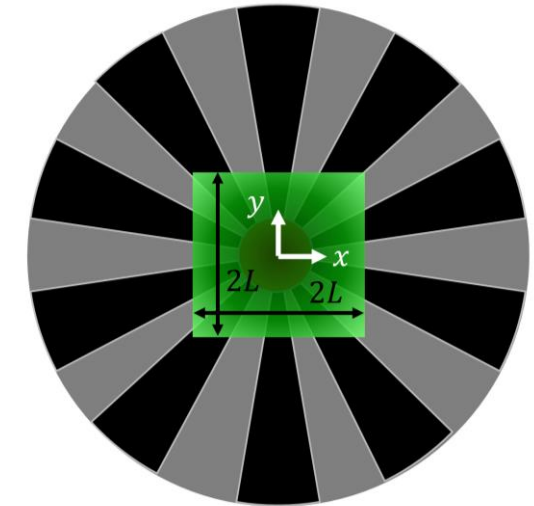
$$F_U(a) = 2\alpha \int_{-\infty}^a \left(\frac{5}{3} - u^2 \right) du$$

$$F_U(a) = 2\alpha \int_{-1}^a \left(\frac{5}{3} - u^2 \right) du$$

$$F_U(a) = 2\alpha \left[\frac{5}{3}u - \frac{u^3}{3} \right]_{-1}^a$$

$$F_U(a) = \frac{2\alpha}{3} \left(5a - \frac{a^3}{3} - \left[\frac{5}{3}(-1) - \frac{(-1)^3}{3} \right] \right)$$

$$\begin{aligned} F_U(a) &= 2\alpha \left(\frac{5}{3}a - \frac{a^3}{3} + \frac{5}{3} - \frac{1}{3} \right) \\ &= \frac{2}{3}\alpha(5a - a^3 + 4) \end{aligned}$$



$$f(u, v) = \alpha(2 - u^2 - v^2)$$

For a valid marginal PDF, $F(\infty) = 1$

$$F_U(1) = \frac{2}{3}\alpha(5 - 1 + 4) = \frac{2}{3}\alpha(8) = 1$$

$$\alpha = 3/16$$



Expected Value for Continuous RVs

The expected value of a continuous joint RV $h(X, Y)$ is given by

$$E[h(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dx dy$$

$$E[h(X, Y)] = \sum_x \sum_y h(x, y) p(x, y)$$

For example, let h be the score, given by

$$h(U, V) = 1 - |U| - |V|$$

The expected value is then

$$E[h(U, V)] = \alpha \int_{-1}^1 \int_{-1}^1 (1 - |u| - |v|)(2 - u^2 - v^2) du dv$$

With the help of Wolfram Alpha

$$E[h(U, V)] = \frac{2}{3} \alpha = \frac{2}{3} \frac{3}{16} = \frac{1}{8}$$

WolframAlpha | PRO FOR EDUCATORS

integrate(integrate A*(1-|u|-|v|)(2-u^2-v^2) du between -1 and 1) dv between -1 and 1

$$\int_{-1}^1 \int_{-1}^1 A(1 - |u| - |v|)(2 - u^2 - v^2) du dv = \frac{2A}{3}$$

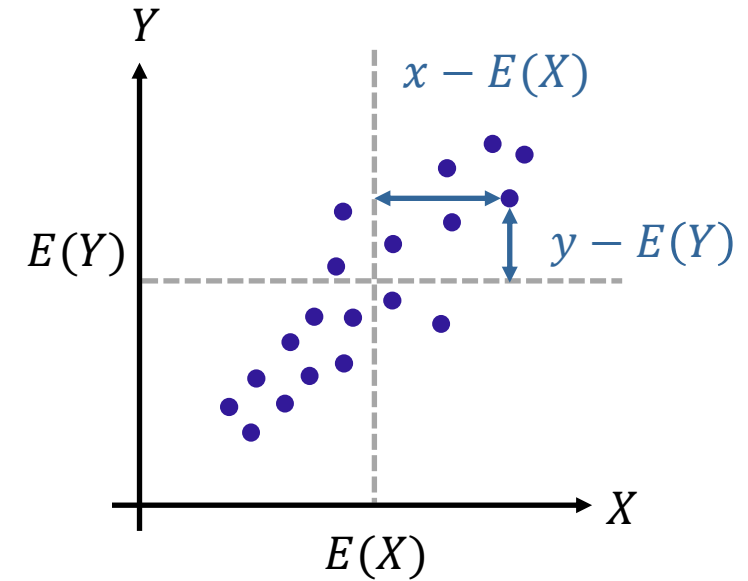


Covariance and Correlation

for Random Variables

Covariance

- When 2 RVs are not independent, we might ask
 - ‘How strongly are they related?’
 - ‘How much variance do they share?’
- We use the covariance
 - the expected value of the product of the deviations from the expected values



$$\text{cov}(X, Y) = E[(x - E(X))(y - E(Y))]$$

$$h(x, y) = (x - E(X))(y - E(Y))$$

$$\text{cov}(X, Y) = \sum_x \sum_y (x - E(X))(y - E(Y))p(x, y)$$

$$E[h(X, Y)] = \sum_x \sum_y h(x, y)p(x, y)$$

Discrete

$$\text{cov}(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - E(X))(y - E(Y))f(x, y) dx dy$$

$$E[h(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dx dy$$

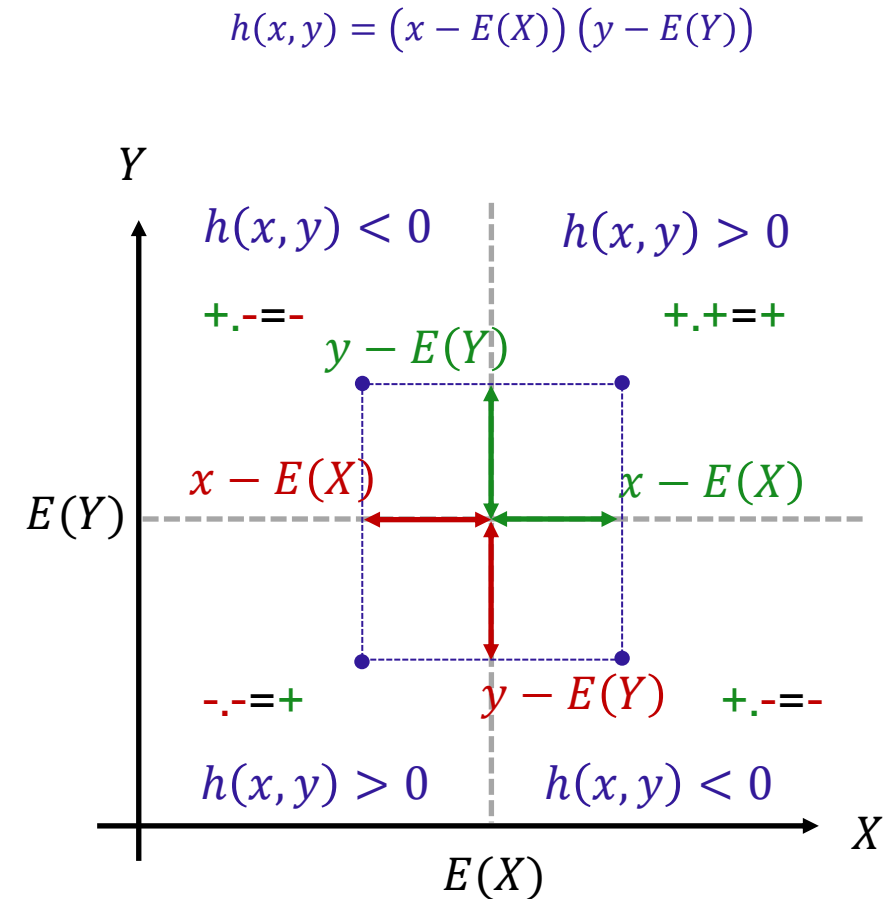
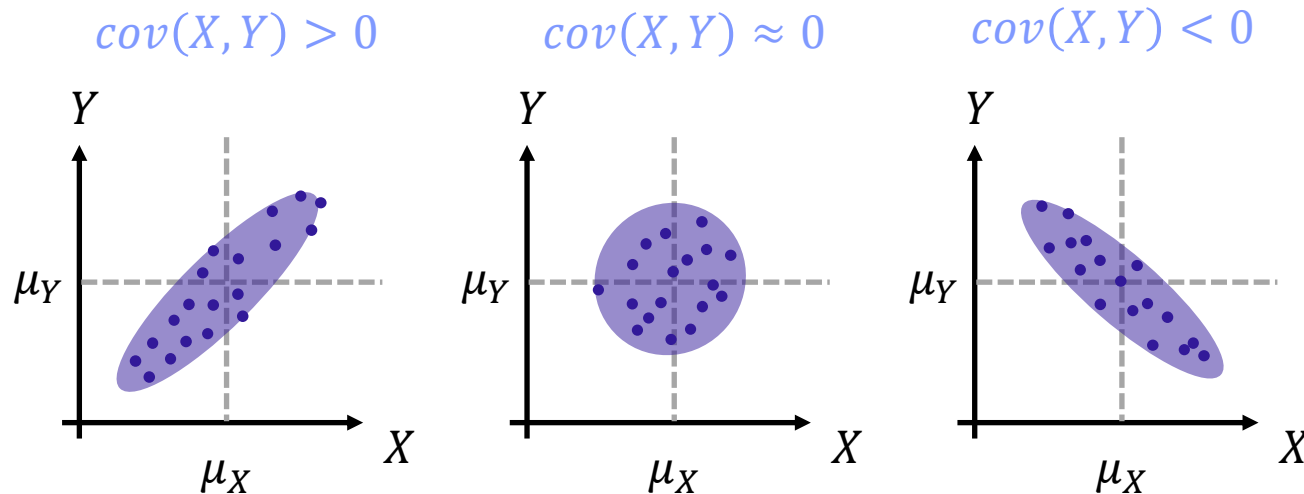
Continuous

Covariance Intuition

$$\text{cov}(X, Y) = E[(x - E(X))(y - E(Y))]$$

$$\text{cov}(X, Y) = \sum_x \sum_y (x - E(X))(y - E(Y))p(x, y)$$

- Note no square for covariance, as we want the information from the signs



Covariance for Tyres and Brakes



$$\text{cov}(X, Y) = \sum_x \sum_y (x - E(X))(y - E(Y))p(x, y)$$

$E(X)$ via marginal PDF

$E(Y)$ via marginal PDF

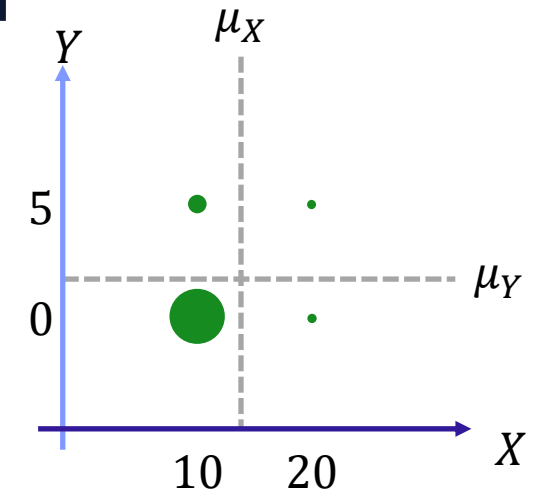
Then covariance

$p(x,y)$		x		
		10	20	
y	5	0.20	0.10	0.3
	0	0.60	0.10	0.7
p_X		0.8	0.2	
				p_Y

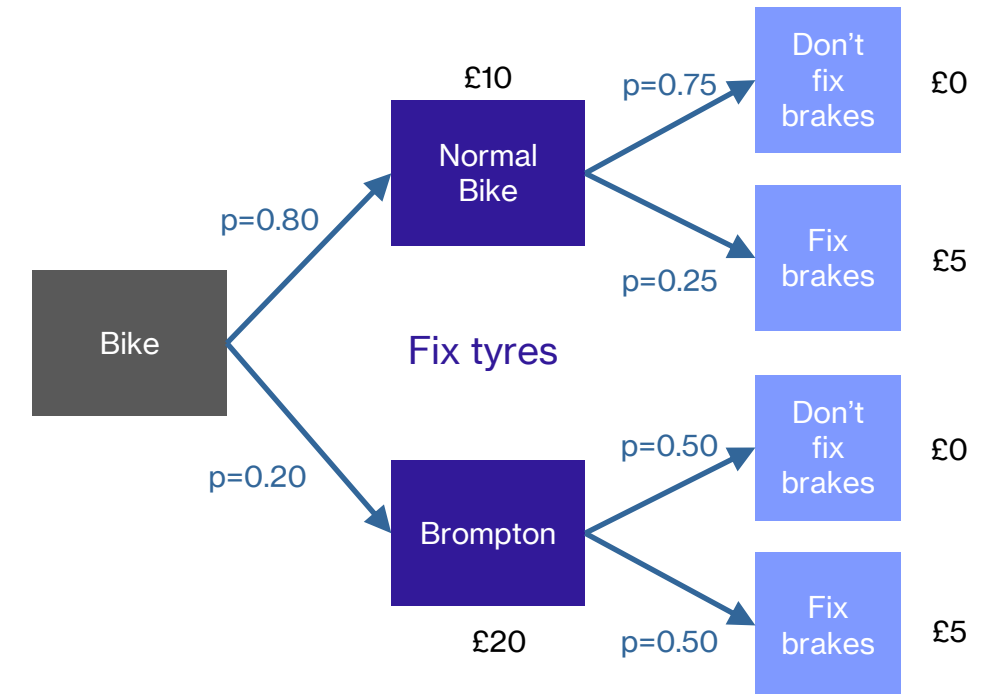
Covariance Interpretation



$$\text{cov}(X, Y) = 2$$



- Positive covariance
- Implies that if we earn more from the tyres than the average* we're also more likely to earn more than the average from the brakes
- But what is the relevance of 2
- What would be implied by 0.2 or 20?
- We need to normalise the covariance
- We need the correlation coefficient



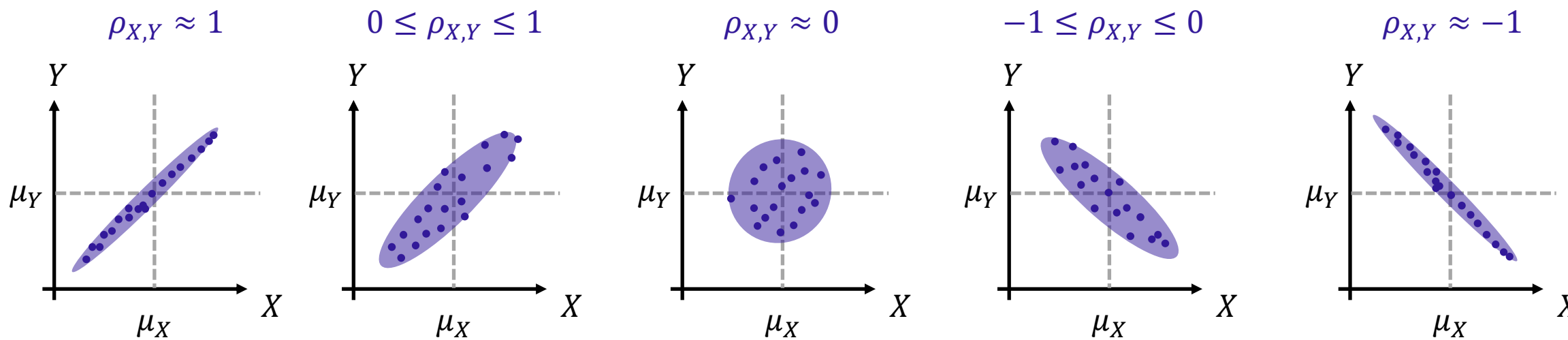
*which is the case for the Brompton, which earns us £20 compared to the expected value of £12

Population Correlation Coefficient, ρ

- Correlation coefficient is a normalised covariance:

$$\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sqrt{V(X)V(Y)}}$$

with $-1 \leq \rho_{X,Y} \leq 1$



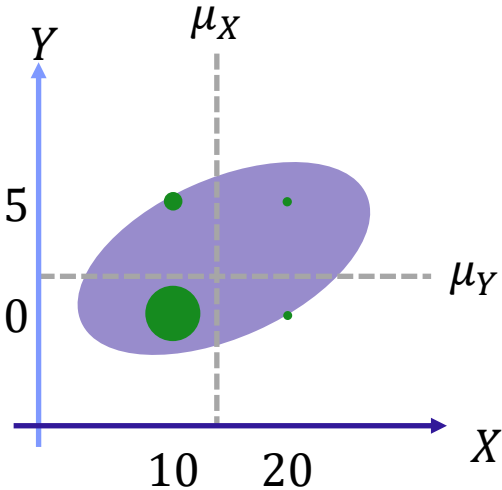
Correlation Coefficient – Tyres and Brakes



$$\rho_{X,Y} = \frac{cov(X,Y)}{\sqrt{V(X)V(Y)}}$$

$$V(X) = \sum_x (x - E(X))^2 p_X(x)$$

$cov(X,Y) = 2$
 $E(X) = 12$
 $E(Y) = 1.5$



$$V(Y) = \sum_y (y - E(Y))^2 p_Y(y)$$

- There is a ‘weak’ correlation
- More on this later!

$$\rho_{X,Y} = \frac{cov(X,Y)}{\sqrt{V(X)V(Y)}}$$

p(x,y)		x		
		10	20	
y	5	0.20	0.10	0.3
	0	0.60	0.10	0.7
		p _X	0.8	0.2

Joint Random Variables

What we covered

- Continuous
 - Probability Density Function
 - Cumulative Distribution Function
 - Percentiles
 - Expected Value
 - Variance
- Joint Probability Distributions
 - PMF, PDF and marginals
 - Expected value
 - Independence
- Covariance
- Correlation

Up next

Lecture 5: Normal Distribution, Statistics and Intervals